



■ CS177 Bioinformatics: Software Development and Applications

Lecture 18: Single-cell Multi-omics

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Learning objectives



- At the end of this lecture, you should be able to:
 - Explain the concept and benefits of multi-omics
 - Identify key single-cell multi-omics sequencing technologies
 - Explain the different fusion strategies (early, intermediate, and late fusion) and their applications in integrating multi-omics data
 - Describe how technologies like CITE-seq allow for the simultaneous profiling of different molecular measurements within individual cells
 - Describe how GLUE (graph-linked unified embedding) uses omics-specific variational autoencoders and a prior knowledge-based guidance graph to integrate single-cell multi-omics data
 - Evaluate the performance of multi-omics data integration methods using metrics, e.g. FOSCTTM



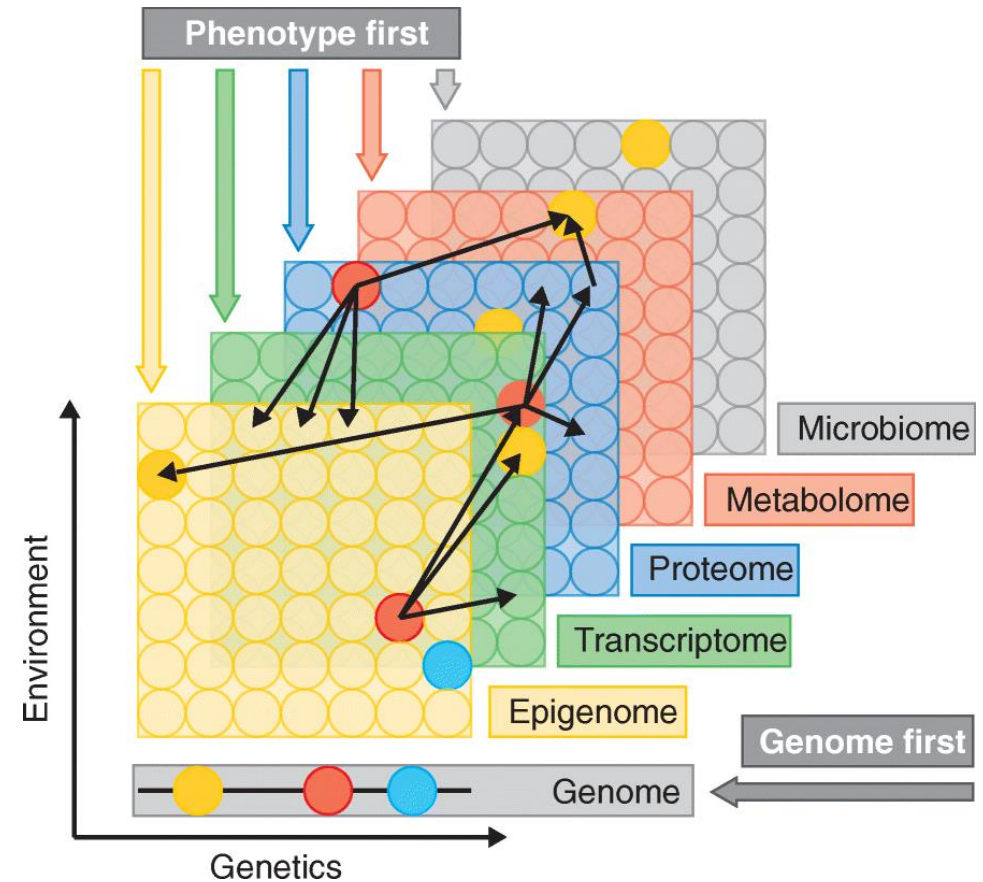


Overview of Single-cell Multi-omics



What is multi-omics?

- Integration of data from multiple omics groups (genomics, transcriptomics, proteomics, epigenomics) from the same biological samples
- **Purpose:**
 - To gain a comprehensive understanding of biological systems by capturing and analyzing multiple layers of cellular activity.
- **Benefits:**
 - Offers a holistic view of the molecular dynamics and interactions within cells.
 - Enhances the accuracy of biological insights by combining multiple data layers.



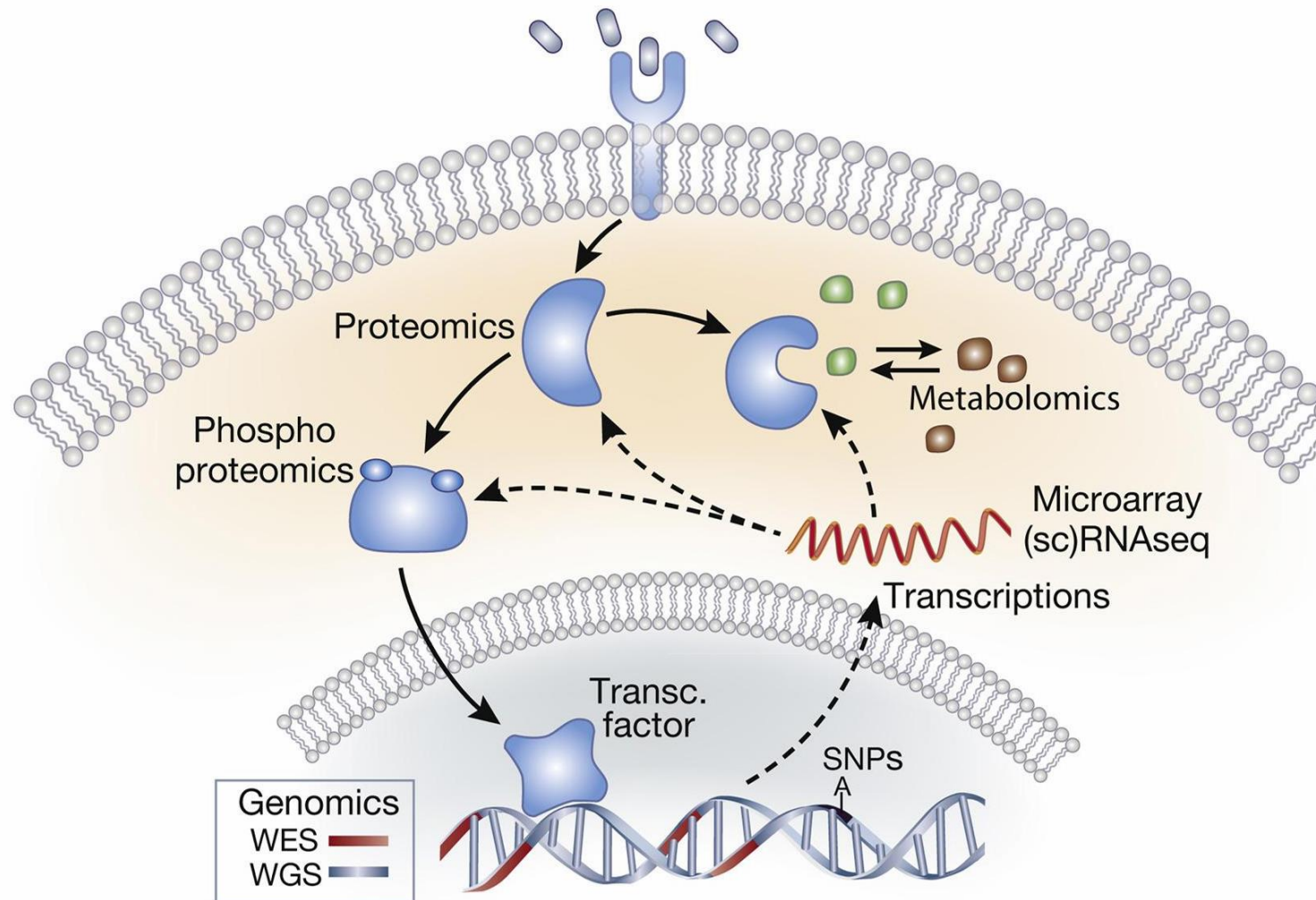
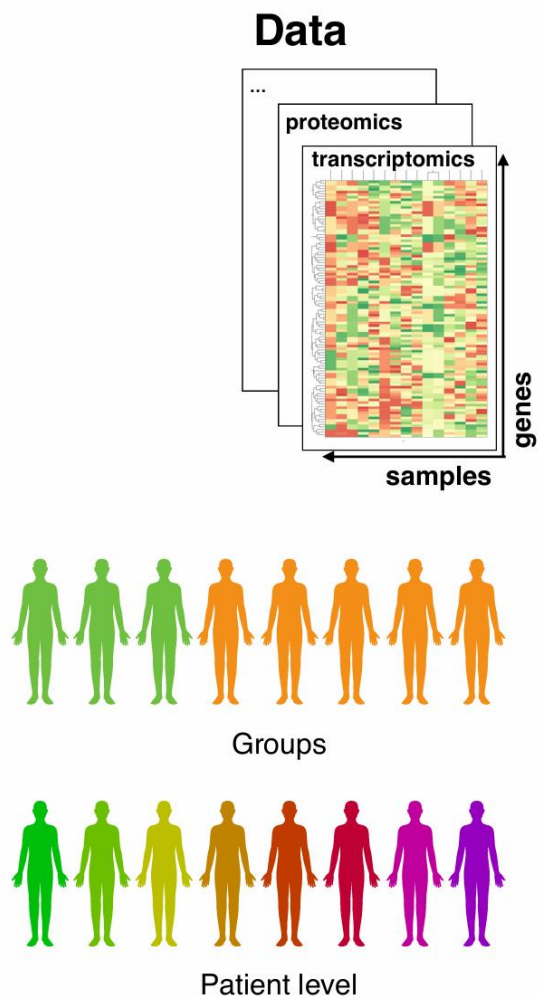
Hasin et al. Genome Biology (2017)
<https://genomebiology.biomedcentral.com/articles/10.1186/s13059-017-1215-1>



Different omics technologies to measure molecular processes



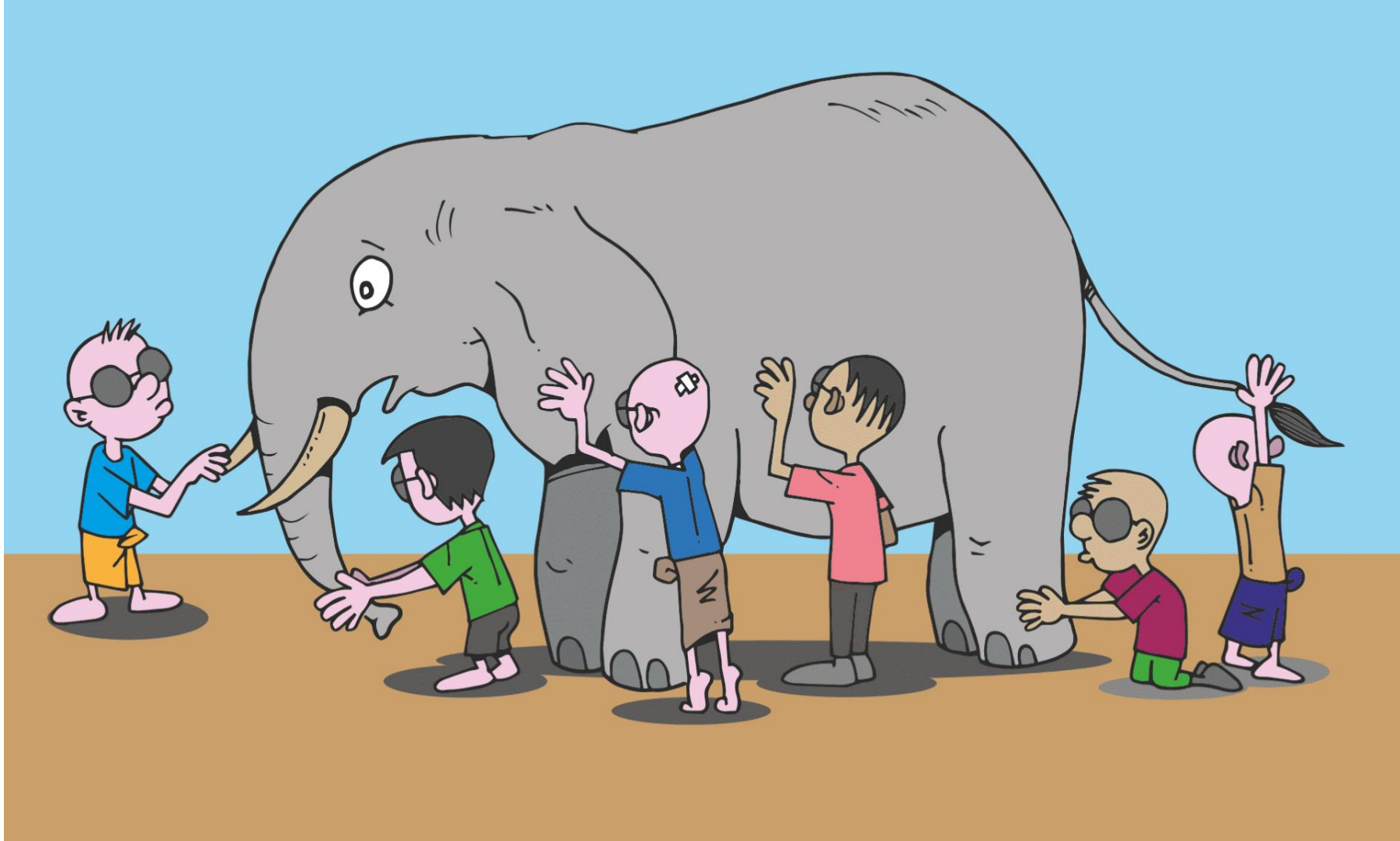
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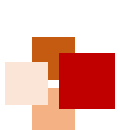
Why need multi-omics?



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From bulk to single-cell multi-omics



- **Bulk Multi-Omics:**

- **Aggregate Data:** Analyzes pooled cells, providing average measurements that may mask cellular diversity.
- **Limitations:** Less effective at detecting minor cell populations or subtle molecular changes within cells.
- **Applications:** Useful for gaining a general overview of molecular signatures across a large number of cells.

- **Single-Cell Multi-Omics:**

- **High Resolution:** Analyzes molecular data from individual cells, revealing cellular heterogeneity.
- **Precision:** Identifies unique cell types and states, capturing rare populations often missed in bulk analysis.
- **Applications:** Crucial for understanding complex biological processes such as tumor microenvironment diversity and stem cell differentiation.

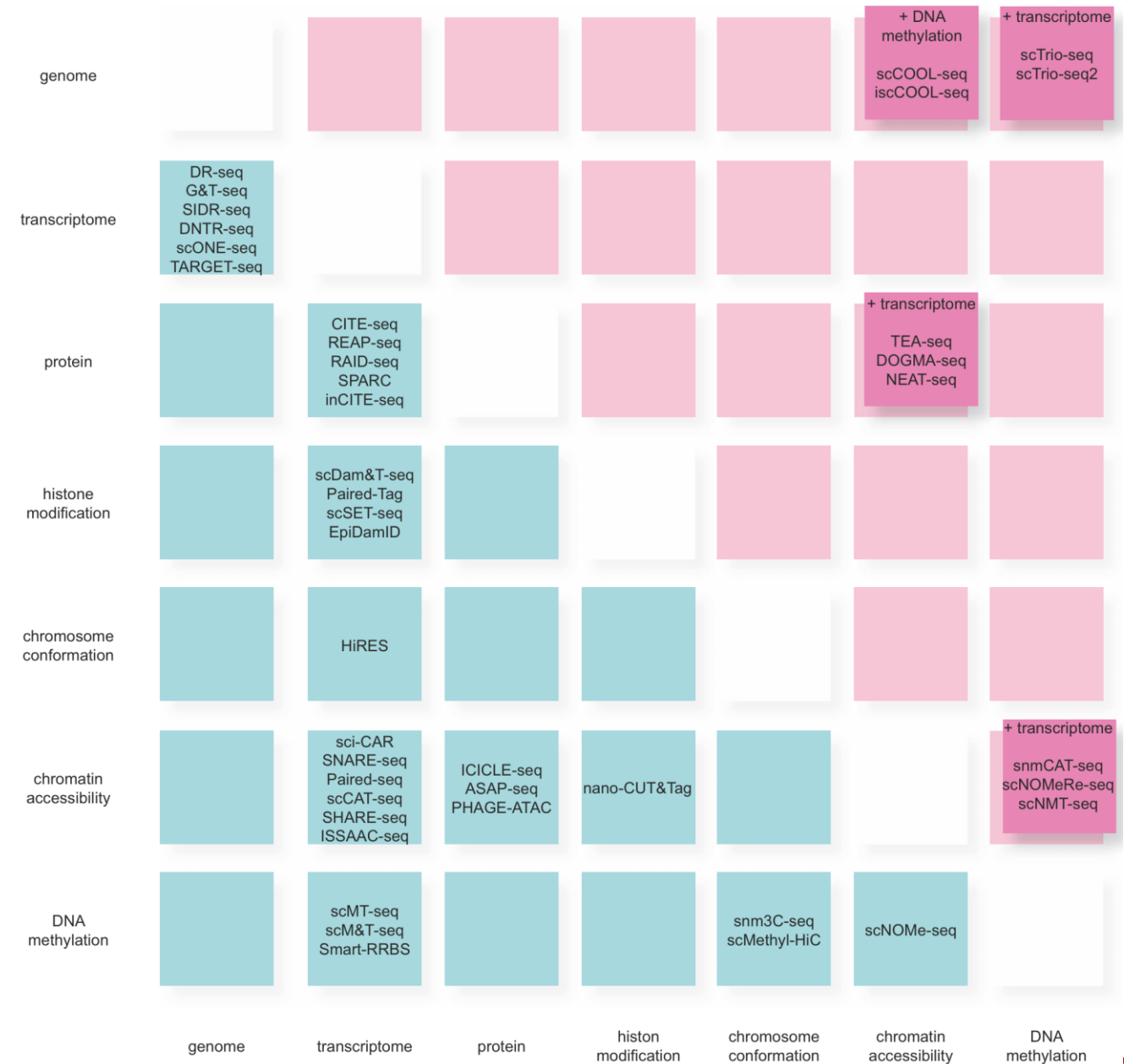


Sequencing methods for single-cell multi-omics analysis



- Numerous single-cell multi-omics sequences have been developed and can be classified into seven types
- The blue box represents dual omics technology
- The magenta box represents technology that handles three or more omics simultaneously

Lim et al. (2024) Experimental & Molecular Medicine 56:515–526;
<https://doi.org/10.1038/s12276-024-01186-2>





Challenges in Single-Cell Multi-Omics



- **Data complexity:**
 - Handling and integrating diverse data types (genomics, proteomics, etc.) that vary significantly in scale, quality, and information content
- **Technical hurdles:**
 - Issues with data sparsity, noise, and variability among single cells, making it difficult to derive accurate conclusions
- **Computational demands:**
 - Need for sophisticated computational tools to manage, process, and analyze **large** datasets effectively
- **Interpretation challenges:**
 - Difficulty in interpreting results due to the complexity of biological systems and the multi-dimensional nature of the data

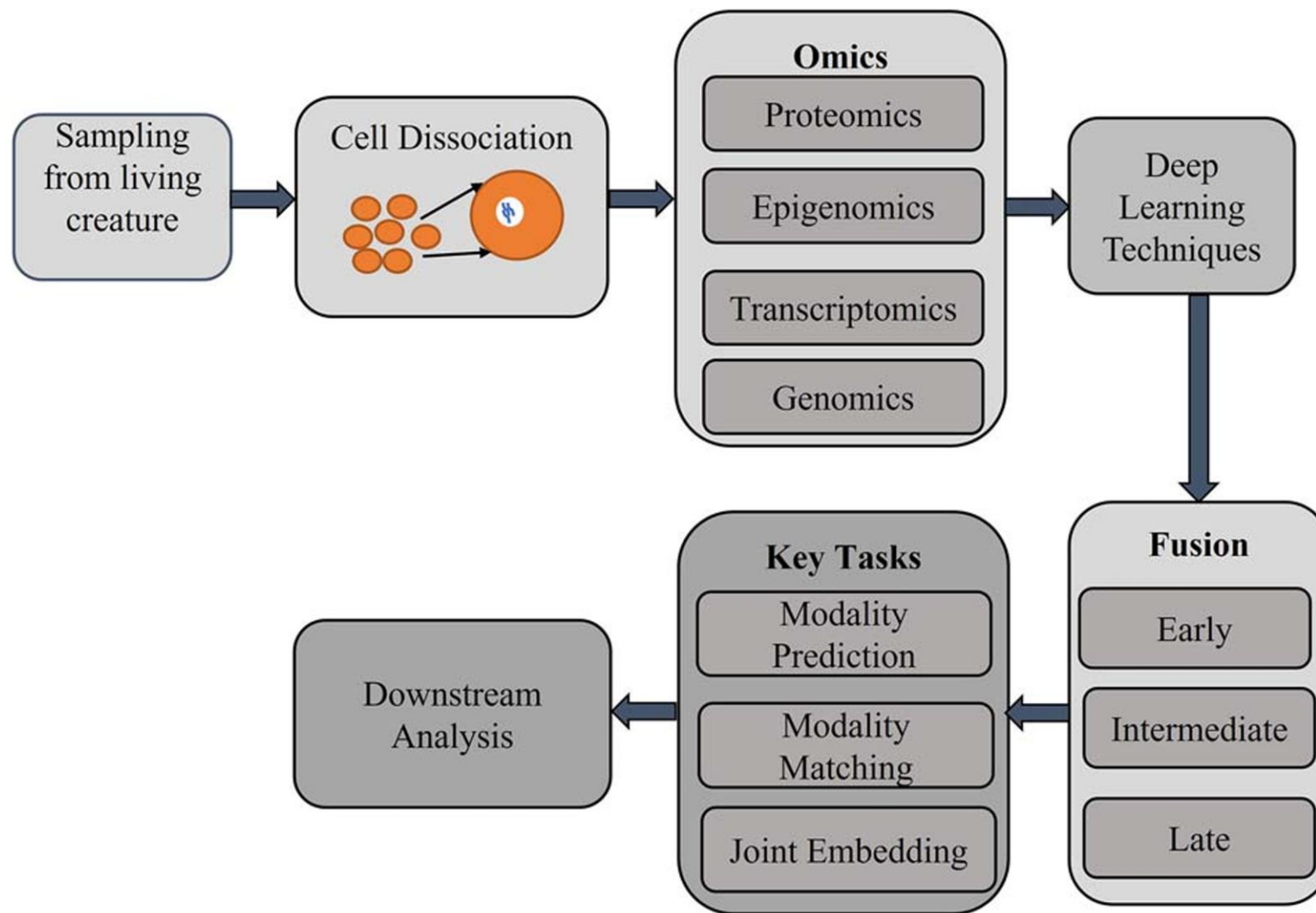


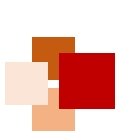


Integration of Multi-omics Data



Workflow of deep learning-based single-cell multi-omics data integration





Fusion strategies

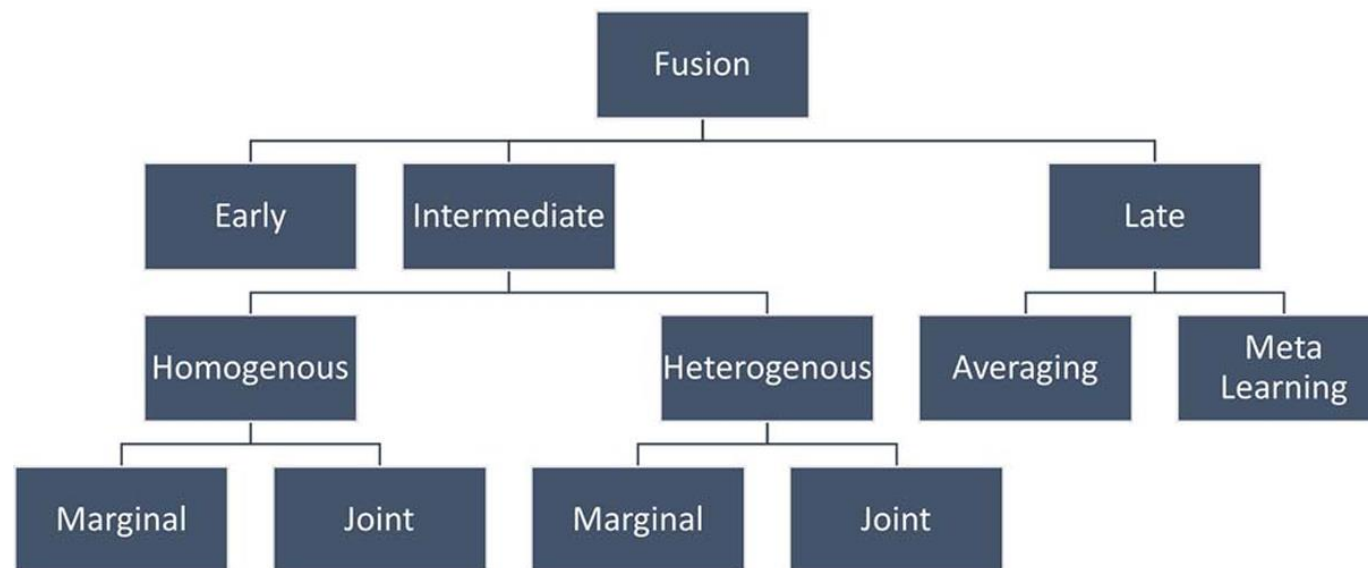
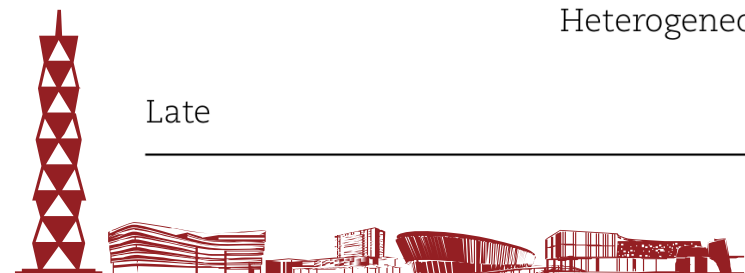
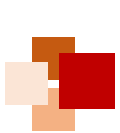


Table 4. Classification of single-cell multimodal studies according to fusion strategy

Fusion	Category 1	Category 2	Papers
Early			scMVAE (with direct learning), totalVI and SCALEX, STACI
Intermediate	Homogeneous	Marginal	MAGAN, SMILE, scMoGNN (with modality prediction and modality matching tasks) and GLUE
		Joint	SCIM, scMM, Cobolt, BABEL, scMVAE, scMoGNN (with joint embedding task), scMVP, scJoint, scDART and Portal
	Heterogeneous	Marginal	–
		Joint	Crossmodal-AE, SAILERX and sciCAN
Late			k-Coupled AE and MIRA

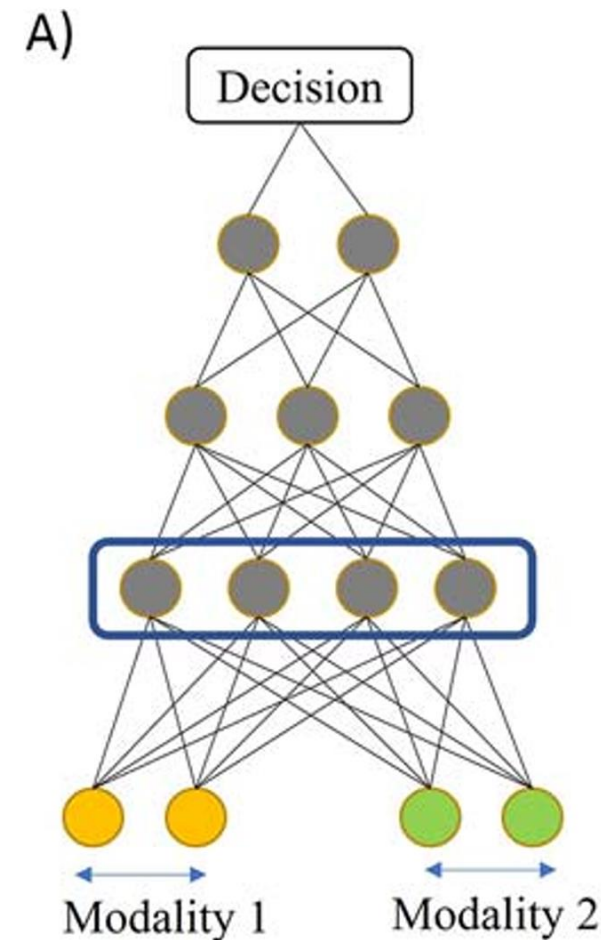




Early fusion



- **Method:** Involves concatenating input features from different modalities to serve as the input of a deep learning model.
- **Examples:** scMVAE, totalVI, STACI, and SCALEX use this strategy.
- **Process:** These models aggregate data from various modalities first and then utilize Variational Autoencoders (VAEs) to learn a joint latent embedding.



Intermediate fusion



- **Method:**

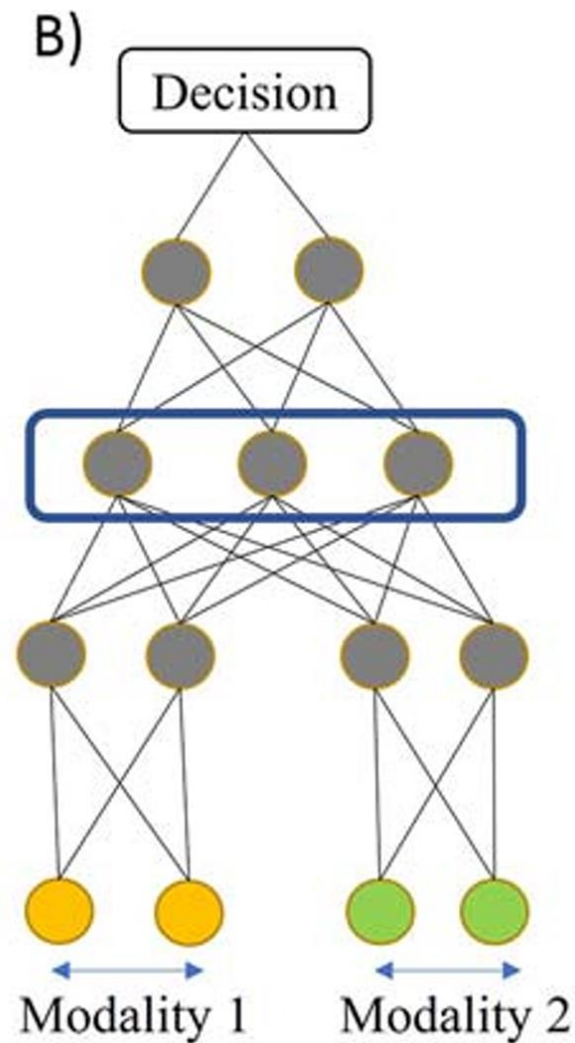
- Modalities are learned separately first and then combined within the model.
- Allows for initial independent processing of each data type, followed by a later stage of integration.

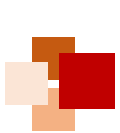
- **Categories of Intermediate Fusion:**

- **Homogeneous:** Same type of neural network is used for learning from each modality.
- **Heterogeneous:** Different types of neural networks are used for different modalities.

- **Representation Types:**

- **Marginal:** Features represent latent components based on a single modality.
- **Joint:** Encodes information from multiple modalities together.





Late fusion



- **Method:**

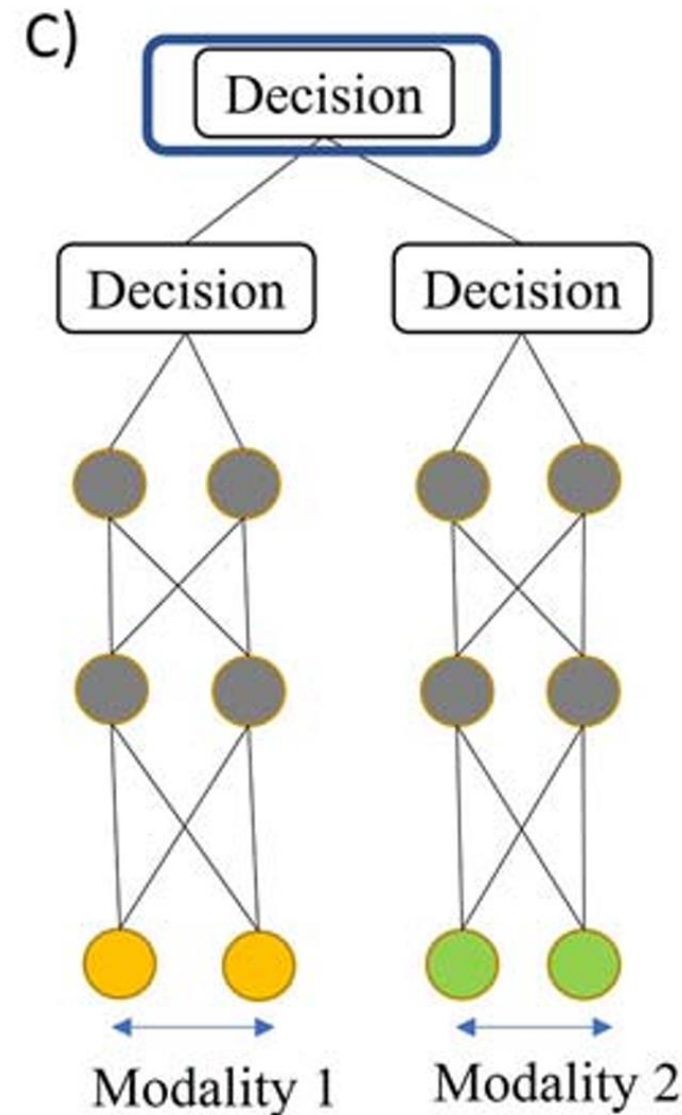
- Integrates decisions from separate models to form a final combined output.
- Each modality is processed independently before final integration.

- **Examples:**

- **k-Coupled AE:** Uses coupled autoencoders (AE) for each modality, later aggregating the latent representations.
- **MIRA:** Utilizes separate variational autoencoders (VAEs) for each modality, combining their outputs at the end.

- **Advantages:**

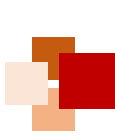
- Maintains the integrity and specificity of each modality's data until the final decision stage.
- Reduces the risk of early overfitting to specific modalities by delaying integration.



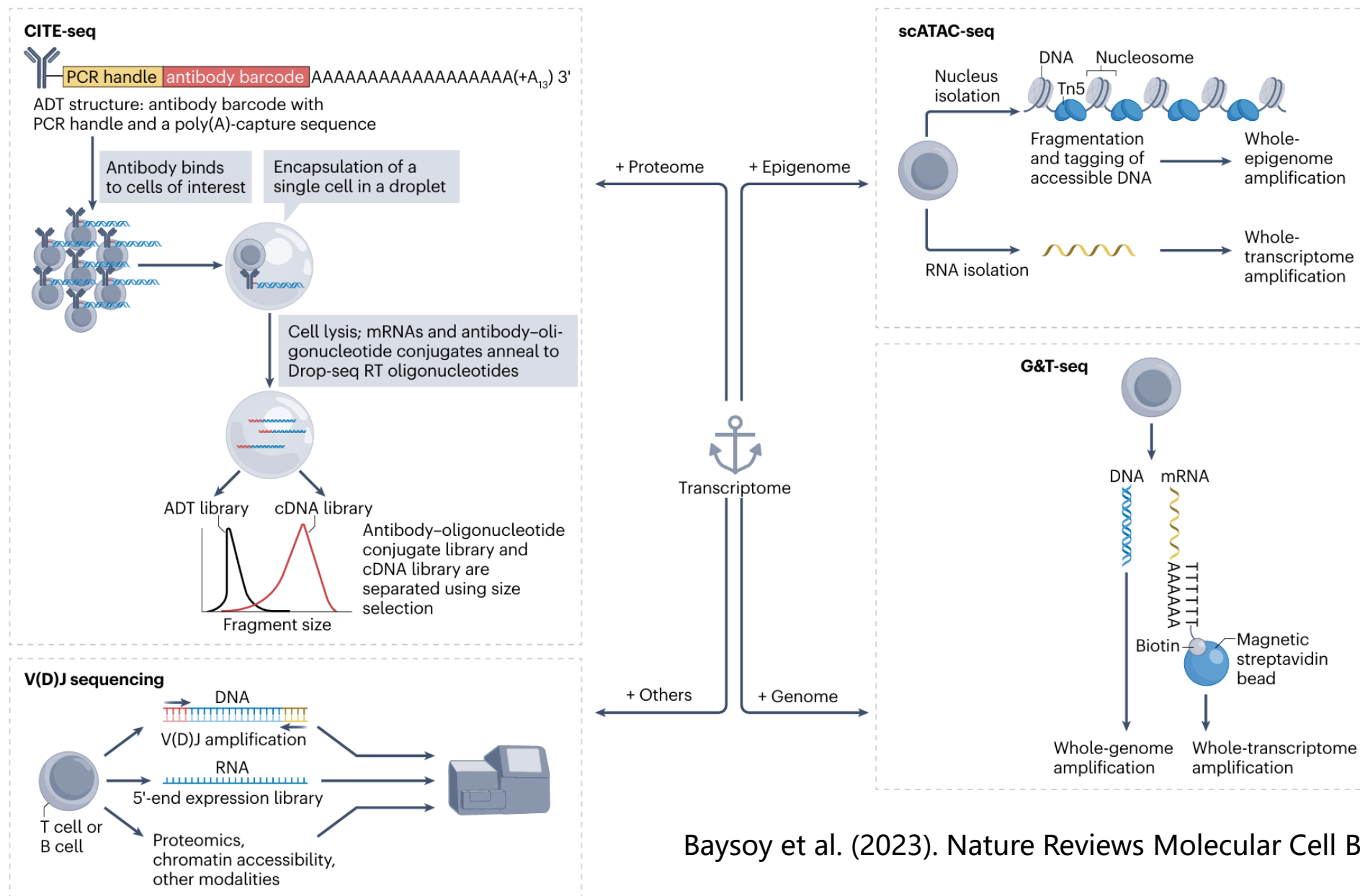


Joint Profiling of Multi-omics from the Same Cells





The landscape of multi-omics sequencing



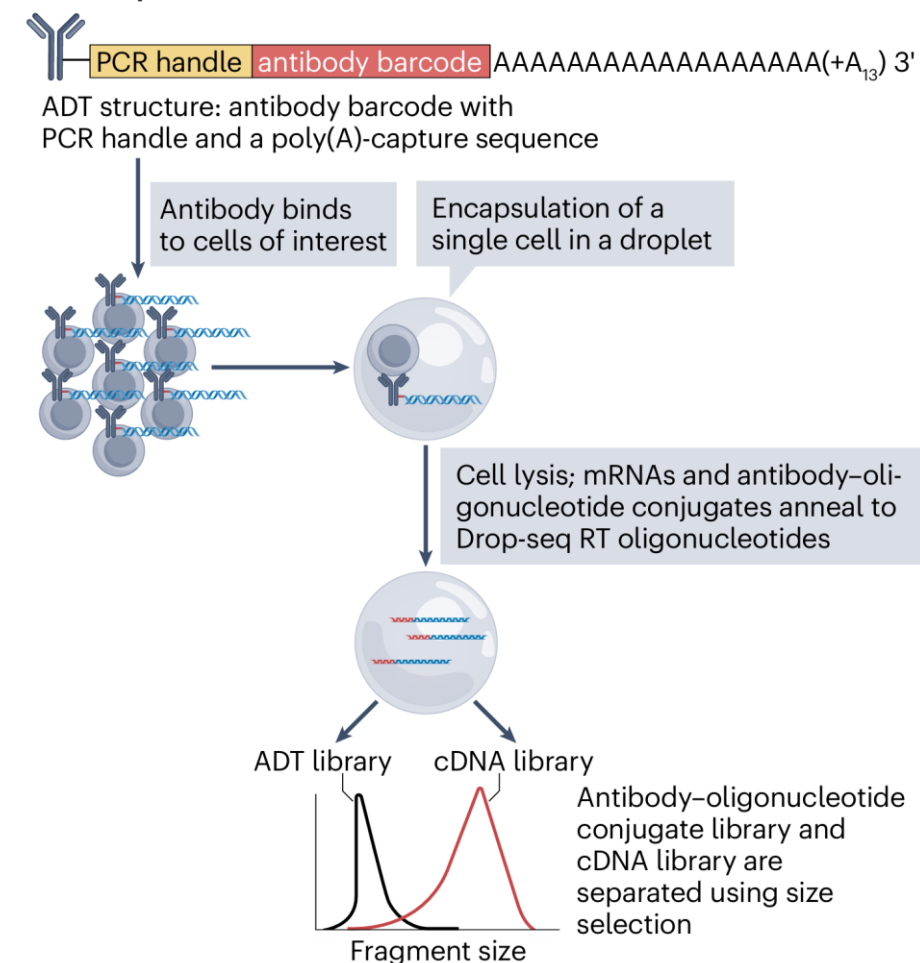
Baysoy et al. (2023). Nature Reviews Molecular Cell Biology

CITE-seq (Cellular Indexing of Transcriptomes and Epitopes by Sequencing)



- CITE-seq is a novel method that integrates cellular protein and transcriptome measurements into a single-cell readout
 - It uses oligonucleotide-labeled antibodies to simultaneously profile protein epitopes and RNA transcripts in individual cells.
 - Antibody-oligonucleotide conjugates are used to tag cell surface proteins.
 - Following cell lysis, both mRNA and antibody-derived tags (ADTs) are captured and sequenced, allowing for the co-analysis of protein expression and transcriptomes.
- **Strengths:**
 - Combines highly multiplexed protein marker detection with unbiased transcriptome profiling.
 - Enables detailed characterization of cellular phenotypes.
 - Offers a more comprehensive view compared to transcriptome measurements alone.
 - **Compatibility and scalability:** CITE-seq is designed to be compatible with existing scRNA-seq platforms and can easily scale with throughput increases, making it adaptable for various research applications.

CITE-seq



Baysoy et al. (2023). Nature Reviews Molecular Cell Biology

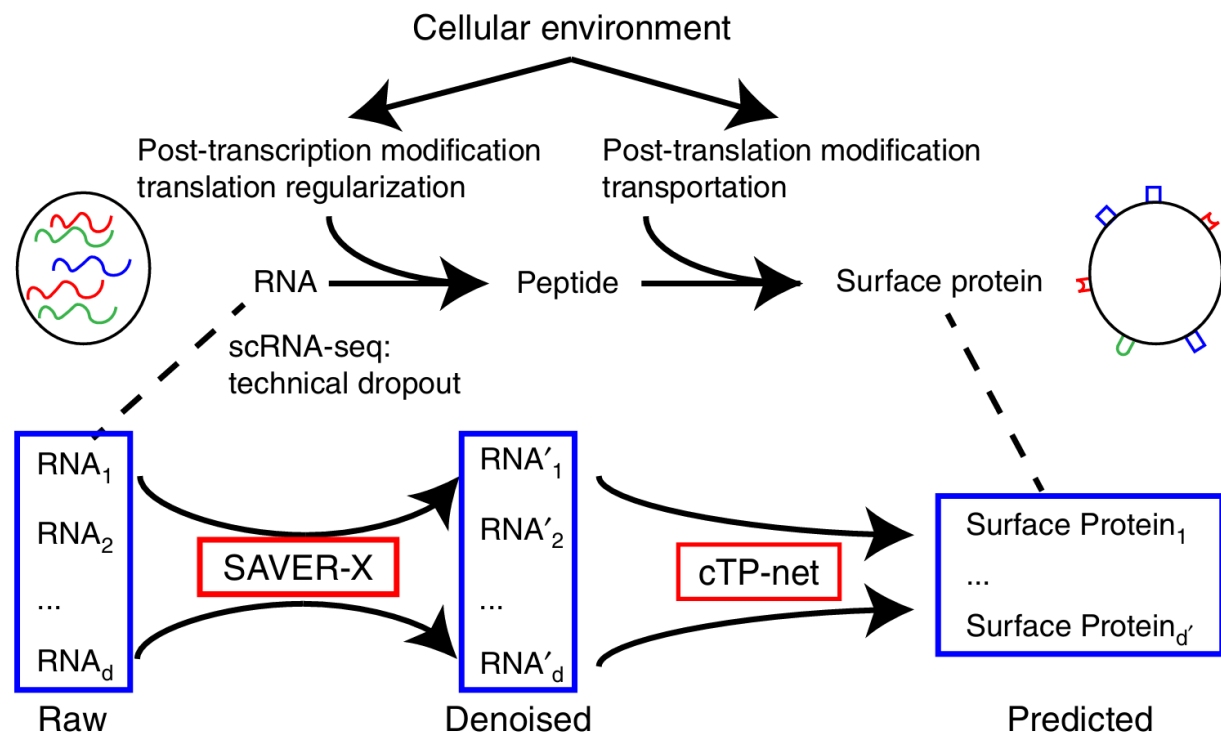
- **Background:**

- **Importance of surface proteins:** Surface proteins are crucial for understanding cellular function and are primary targets for therapeutic intervention.
- **Challenges in direct measurement:** Due to technological and cost limitations, most single-cell studies quantify the transcriptome without measuring corresponding surface proteins.

- **Method overview:**

- **cTP-net (cell Transcriptome to Protein prediction with deep neural network)** is a **transfer learning** approach that uses deep neural networks to predict cell surface protein abundances from single-cell RNA sequencing (scRNA-seq) data.
- It is designed to leverage existing datasets that include both transcriptome and surface protein measurements (e.g. CITE-seq), enhancing the imputation of protein levels from scRNA-seq data.





- The pipeline of cTP-net:

- Step 1: Denoising scRNA-seq Data:** cTP-net first denoises the scRNA-seq count matrix using **SAVER-X**, which improves the correlation between RNA transcript abundances and their corresponding protein measurements
- Step 2: Protein Imputation:** Following the denoising, cTP-net employs a **multiple branch deep neural network (MB-DNN)** to map the denoised transcriptome data to the relative abundance of surface proteins





Case Study: GLUE





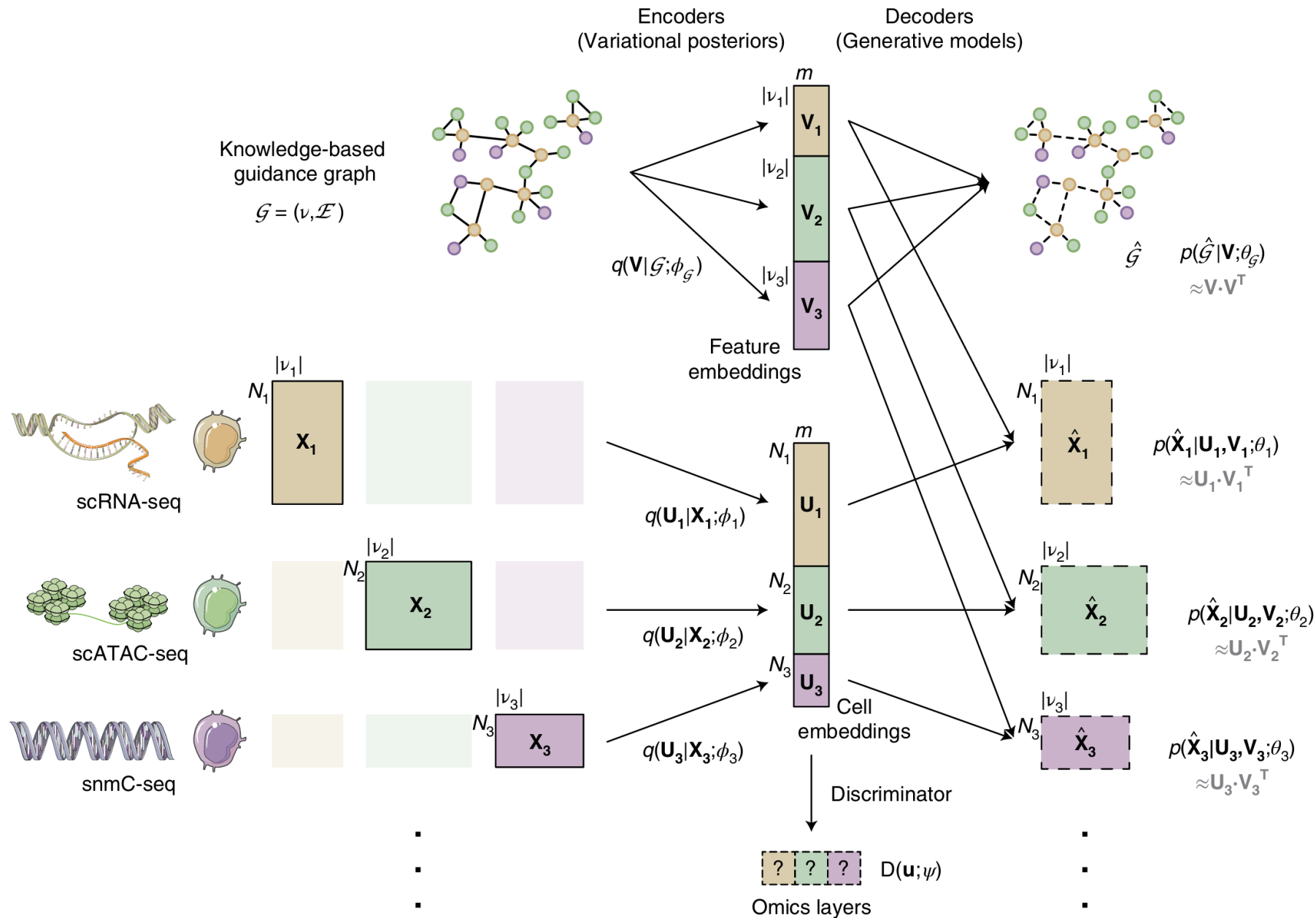
Computational challenges



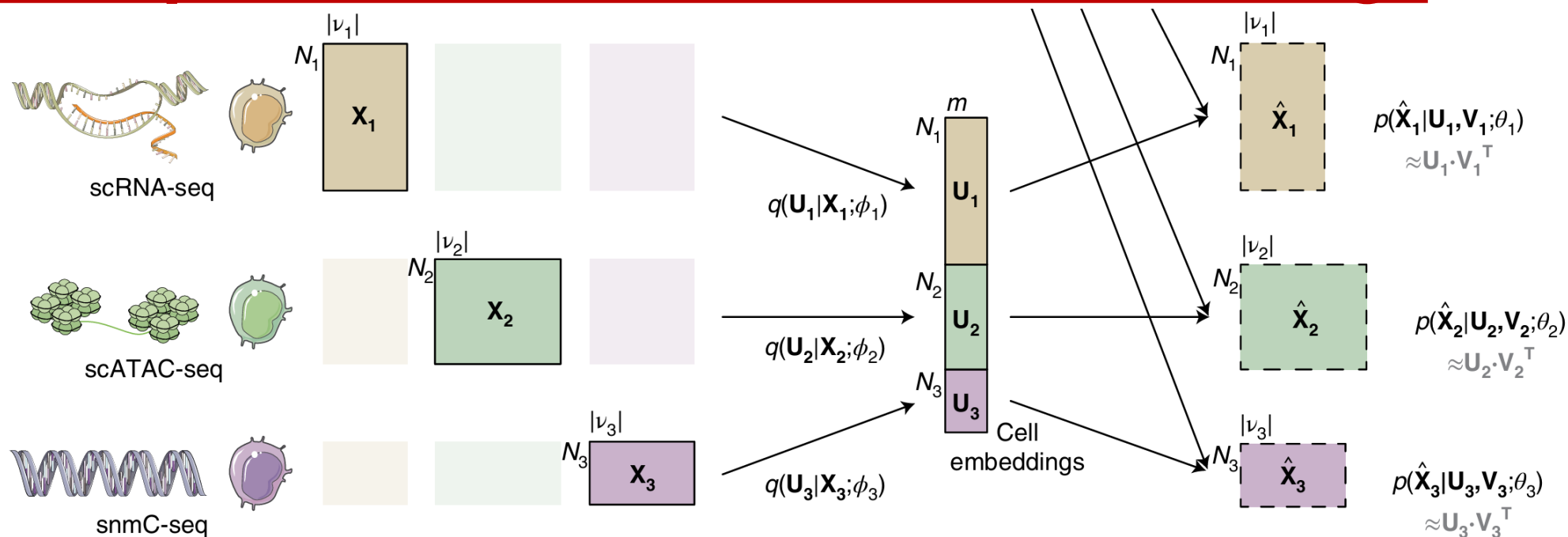
- Integrating unpaired multi-omics data (i.e. diagonal integration)
 - Simultaneous (joint) assays have emerged
 - But usually different omics are measured **independently** and produce **unpaired** data
- **Challenge 1:** Distinct feature spaces of different modalities
 - Different single-cell omics layers, such as scATAC-seq, DNA methylation sequencing (snmC-seq, sci-MET), and the transcriptome (scRNA-seq), measure different types of molecular information and have distinct feature spaces. This makes direct integration difficult
- **Challenge 2:** Scalability
 - With the increasing volume of single-cell data being generated, current integration methods need to be scalable to handle datasets of millions of cells



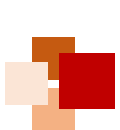
Overview of GLUE (graph-linked unified embedding)



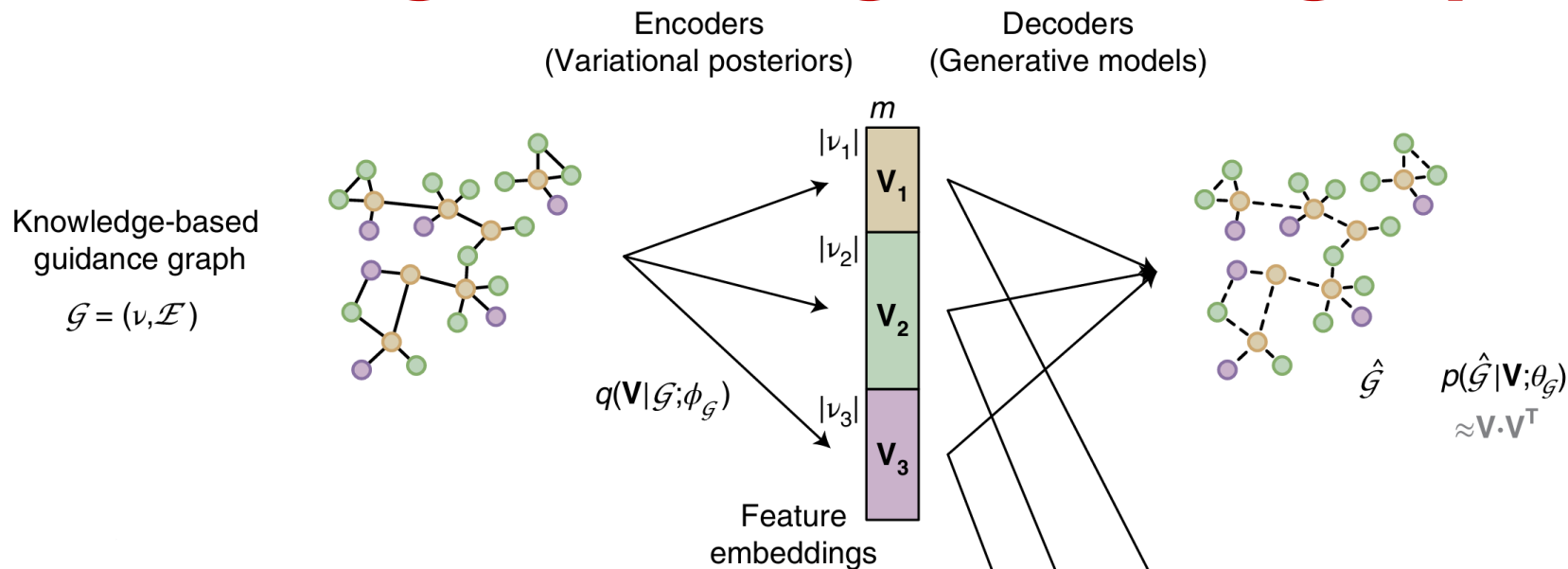
Omics-specific VAEs to learn cell embeddings



- GLUE employs **variational autoencoders (VAEs)** tailored to each omics layer to learn low-dimensional cell embeddings, taking into account the unique biological nature and assay technology of each data type
- **Input:** raw data from particular omics layer
 - Transcriptomics layer: scRNA-seq
 - Chromatin accessibility layer: scATAC-seq
 - DNA methylation layer: snmC-seq
- **Output:** a set of low-dimensional cell embeddings

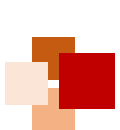


Prior knowledge-based guidance graph

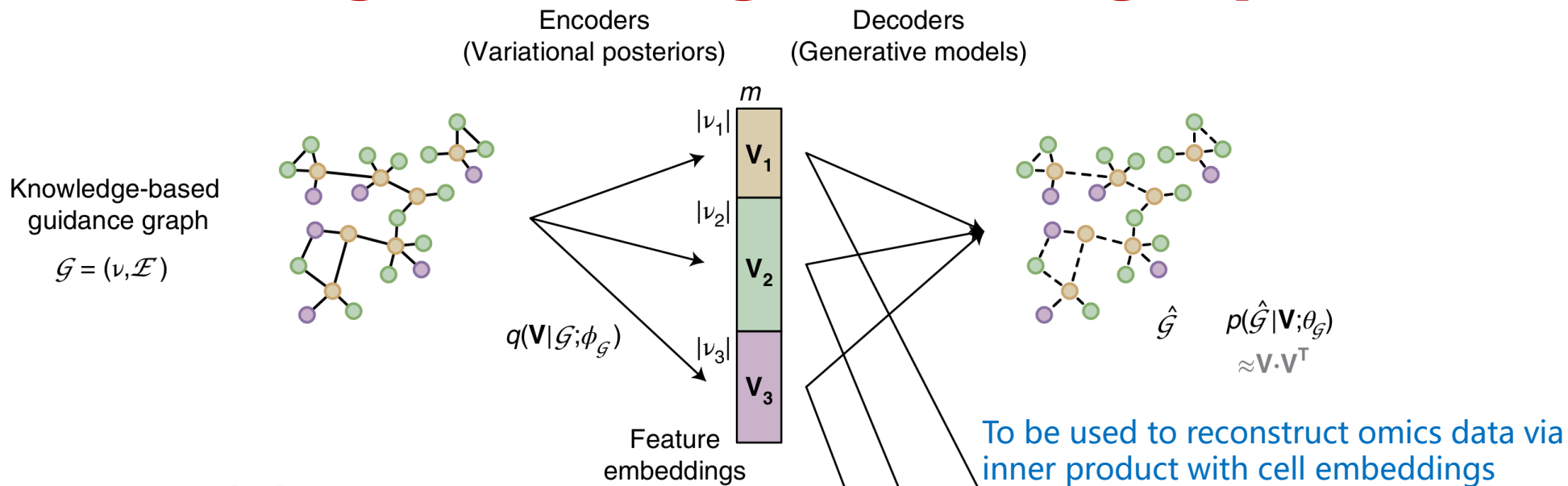


- **Guidance graph construction:** The guidance graph explicitly models cross-layer regulatory interactions to link feature spaces of different omics layers. It incorporates prior biological knowledge where:
 - **Vertices** represent features of various omics layers, e.g., in integrating scRNA-seq and scATAC-seq data, the vertices are genes (from scRNA-seq) and accessible chromatin regions (from scATAC-seq)
 - **Edges** denote signed regulatory interactions, e.g. a positive edge between an accessible chromatin region and its downstream gene, reflecting the regulatory influence of the region on gene expression





Prior knowledge-based guidance graph



- A **graph variational autoencoder** is used to learn feature embeddings

$$\mathbf{V} = (\mathbf{V}_1^T, \mathbf{V}_2^T, \mathbf{V}_3^T)^T$$

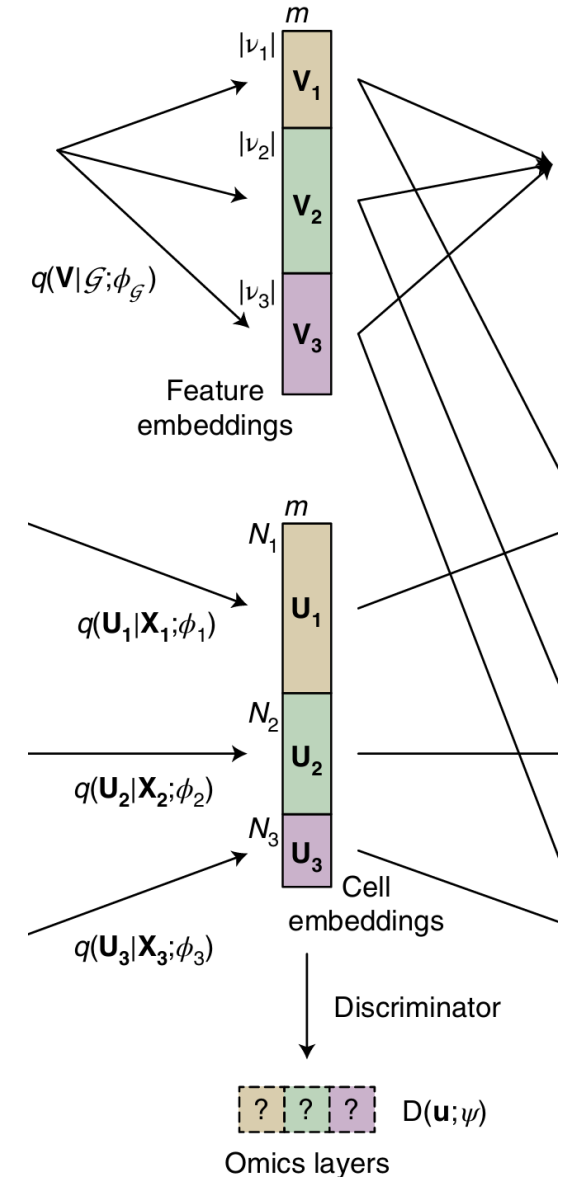
from the prior knowledge-based guidance graph

- The feature embeddings are then used in data decoders
 - To reconstruct omics data via **inner product** with cell embeddings
 - Effectively linking the omics-specific data spaces to ensure a consistent embedding orientation



Adversarial alignment

- Cells are aligned via an **adversarial process** guided by feature embeddings from the guidance graph, using the following steps:
- Discriminator Setup:** A discriminator with a softmax output predicts the omics layer of cells based on their embeddings
- Training the Discriminator:** The discriminator is trained using a cross-entropy loss to minimize classification errors, enhancing its ability to correctly identify the omics layer from the cell embeddings
- Adversarial Training:** The encoders for the cell embeddings are trained to “fool” the discriminator, i.e. adjusted to become indistinguishable by the discriminator, thereby promoting alignment across different omics layers
 - Adversarial training principle:** the encoders improve to counteract the discriminator's classification capabilities
- Iterative Optimization:** Iterative updates to both the discriminator and the encoders
 - First, the discriminator is updated to better classify the embeddings
 - Then the encoders are updated to better confuse the discriminator
 - This iterative process helps refine the embeddings to achieve better alignment across the different omics layers



Systematic benchmarking: biology conservation

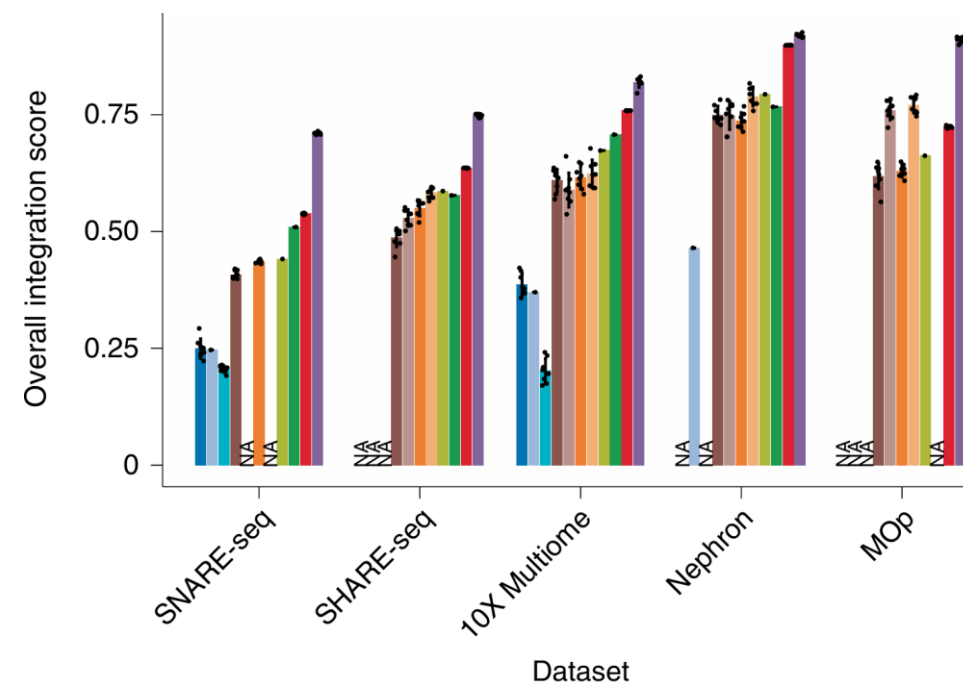
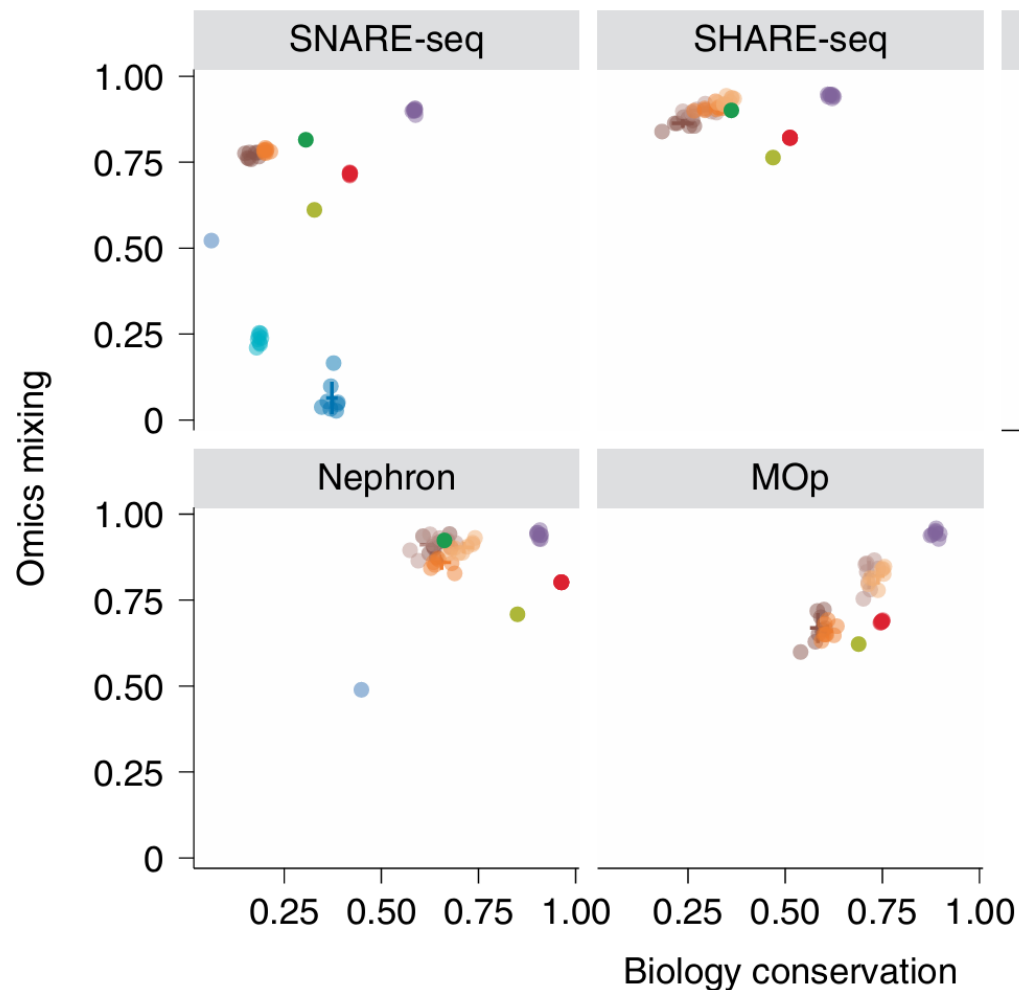


- **Mean Average Precision (MAP):** evaluates the cell type resolution by computing the average precision for each cell type up to each cell type-matched neighbor, with mean average precision being the average of these values across all cells. Higher MAP values indicate better cell type resolution
- **Cell Type Average Silhouette Width (Cell type ASW):** measures how well the cell embeddings of the same type cluster together compared to embeddings of different types. It is calculated as the average silhouette width for each cell type, and higher values suggest better cell type resolution
- **Neighbor Consistency (NC):** assesses the preservation of single-omics data variation after multi-omics integration. It measures the consistency between the nearest neighbors in the single-omics and integrated spaces, with higher values indicating better preservation of data variation
- **Biological conservation:** first conduct min-max scaling for each of the metrics and then compute the average across the three to summarize them into a single metric representing biology conservation:

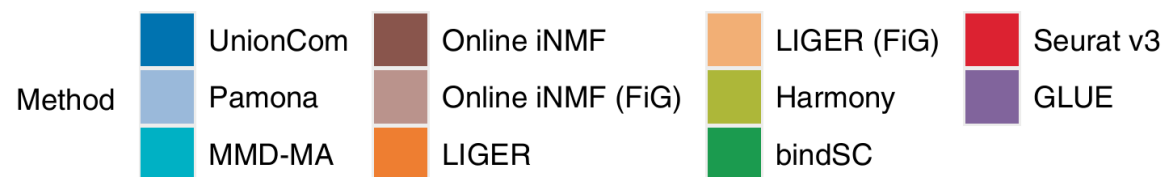
$$\text{biology conservation} = \frac{\text{scale}(\text{MAP}) + \text{scale}(\text{cell type ASW}) + \text{scale}(\text{NC})}{3}$$



Systematic benchmarking: biology conservation and omics mixing



$$\text{overall integration score} = 0.6 \times \text{biology conservation} + 0.4 \times \text{omics mixing}$$



Systematic benchmarking: single-cell level alignment error measured by FOSCTTM



FOSCTTM²⁵ was used to evaluate the single-cell level alignment accuracy. It was computed on two datasets with known cell-to-cell pairings. Suppose that each dataset contains N cells, and that the cells are sorted in the same order, that is, the i th cell in the first dataset is paired with the i th cell in the second dataset. Denote x and y as the cell embeddings of the first and second dataset, respectively. The FOSCTTM is then defined as:

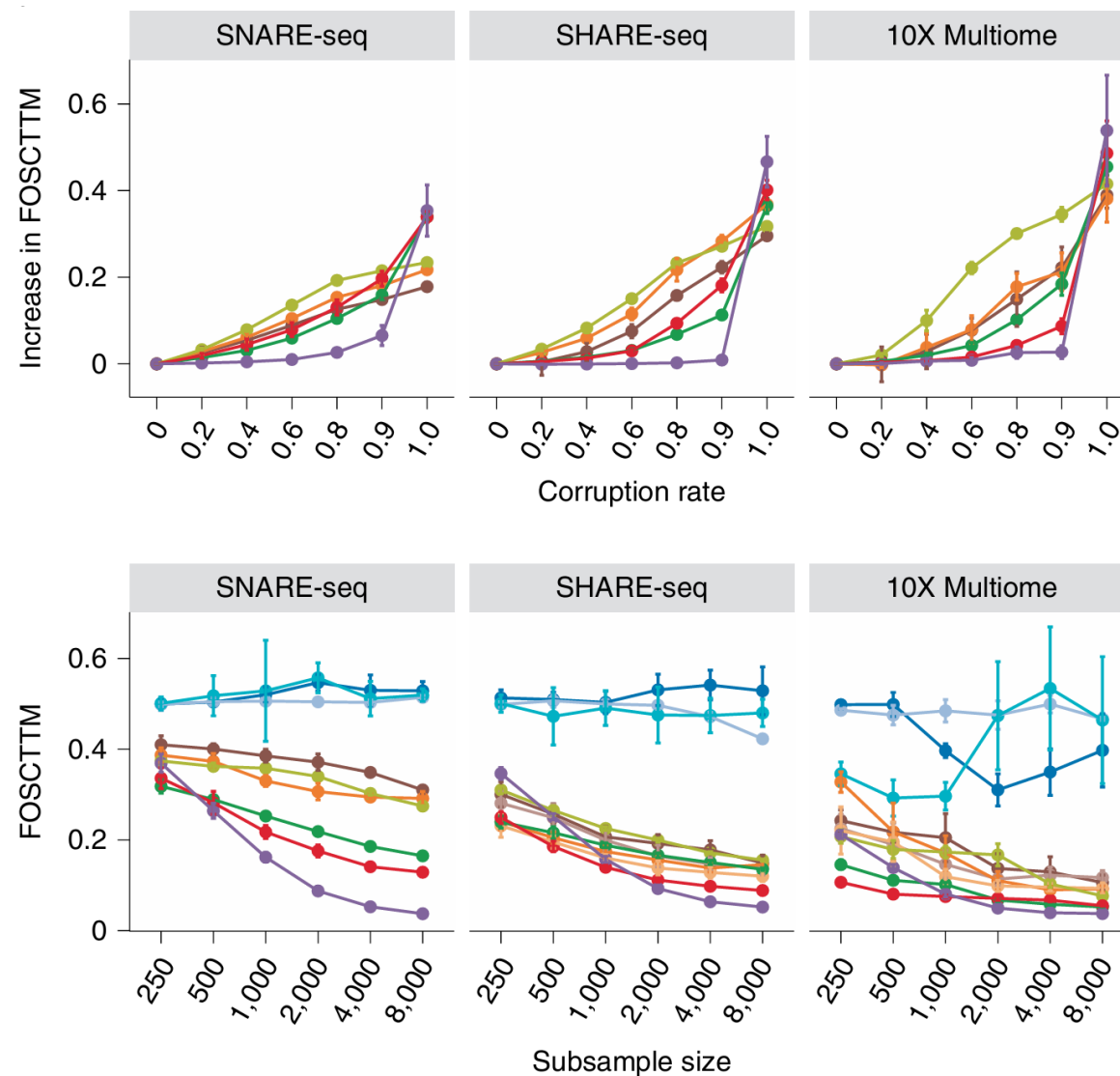
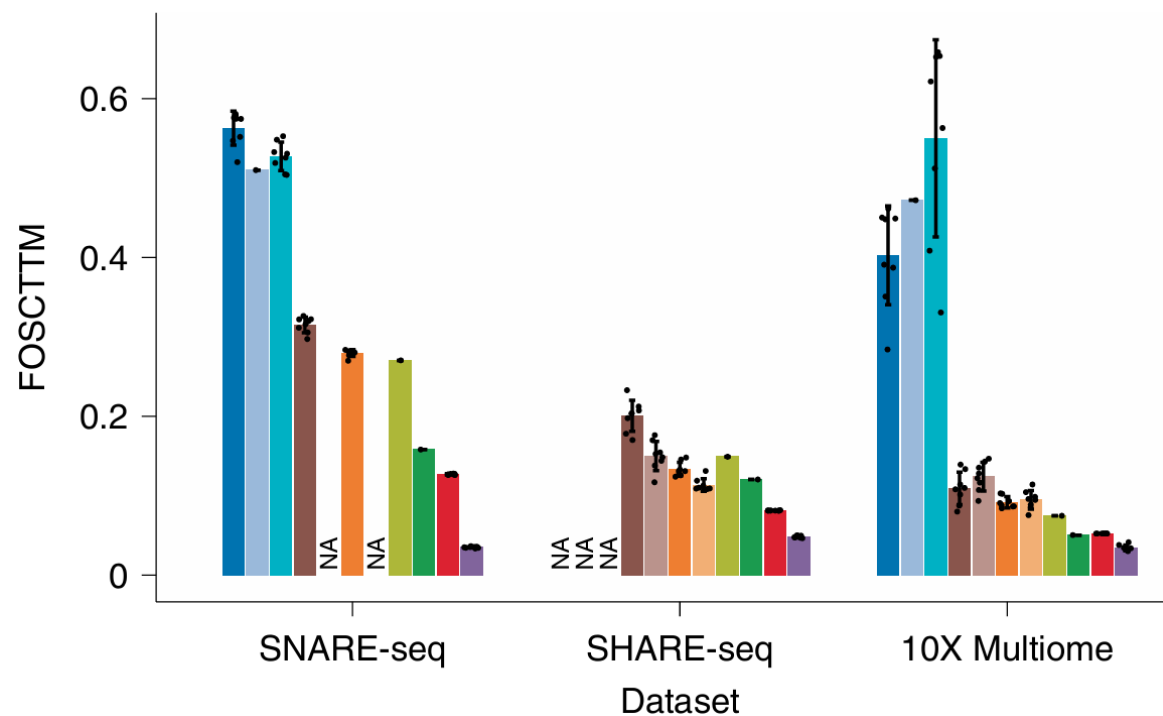
$$\text{FOSCTTM} = \frac{1}{2N} \left(\sum_{i=1}^N \frac{n_1^{(i)}}{N} + \sum_{i=1}^N \frac{n_2^{(i)}}{N} \right) \quad (39)$$

$$n_1^{(i)} = |\{j | d(\mathbf{x}_j, \mathbf{y}_i) < d(\mathbf{x}_i, \mathbf{y}_i)\}| \quad (40)$$

$$n_2^{(i)} = |\{j | d(\mathbf{x}_i, \mathbf{y}_j) < d(\mathbf{x}_i, \mathbf{y}_i)\}| \quad (41)$$

where $n_1^{(i)}$ and $n_2^{(i)}$ are the number of cells in the first and second dataset, respectively, that are closer to the i th cell than their true matches in the opposite dataset. d is the Euclidean distance. FOSCTTM has a range of 0 to 1, and lower values indicate higher accuracy.

Systematic benchmarking: single-cell level alignment error measured by FOSCTTM





Discussion: Modular and generalizable design



- GLUE's modular design allows for easy extension and quick adoption to new analysis tasks and omics data types
 - Independent configuration of data and graph autoencoders, providing flexibility in adapting to various omics data types
 - Incorporate different generative models within the autoencoders to accommodate the specific distribution of each omics layer
- Graph extension and enhancement:
 - The guidance graphs used in GLUE are not limited to multipartite graphs and can embody more complex regulatory patterns, including intra-modality interactions and multi-relations
 - Advanced graph neural network architectures, including hypergraph modeling, can be integrated into GLUE to extract richer information from the regulatory graph
- Generalizability:
 - The framework is not restricted to the integration of specific omics data types but is extensible to new data modalities, such as protein abundance and histone modification data
 - It could be applicable to cross-species data integration, especially when one-to-one orthologs are limited, by compiling orthologs into a guidance graph
 - While primarily designed for unpaired data, GLUE can be extended to leverage paired cell information, such as that obtained from simultaneous multi-omics protocols, to improve alignment





References



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